

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-00-039

WASTE DISCHARGE REQUIREMENTS
FOR
MADISON COMMUNITY SERVICES DISTRICT
MADISON, YOLO COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Madison Community Services District (hereafter Discharger) owns and operates a domestic wastewater collection, treatment and disposal system for discharges from the community of Madison in Yolo County. The treatment and disposal facility consists of seven stabilization ponds.
2. The Board, on 8 December 1989, adopted Waste Discharge Requirements (WDRs) Order No. 89-235 for the facility. Order No. 89-235 is neither adequate nor consistent with current plans and policies of the Board and does not reflect the recent expansion of the facility.
3. Prior to the most recent upgrade, five serially operated stabilization ponds relied on percolation and evaporation to dispose of the community's wastewater. Historically during the winter months, wastewater volume had been known to exceed the design elevation and freeboard in the ponds. The Discharger submitted a Capacity Analysis and Wastewater Management Plan on 21 August 1992, indicating average dry weather flows of 0.129 million gallons per day (mgd), more than twice the limit of 0.047 mgd imposed by Order No. 89-235.
4. The approximate population of Madison is 505, with an estimated daily discharge of approximately 69,000 gpd. The Madison Migrant Center (Center), east of the community of Madison, includes 88 dwelling units. Only occupied during the six-month harvest season of May through October, the legal occupancy limit for the Center is 484 persons with an estimated wastewater discharge of 66,308 gpd.
5. On 25 February 1994, the Board issued Cease and Desist Order (C&D Order) No. 94-062 to the Discharger due to waste discharges contrary to Discharge Prohibitions Nos. 1 and 2, and Discharge Specifications Nos. 3, 4, and 6 of Order No. 89-235. This C&D Order required the Discharger to submit a work plan and schedule for expansion beyond the 0.129 mgd to accommodate growth. The Discharger was required to complete a treatment system expansion in accordance with the following compliance schedule:

<u>Task</u>	<u>Compliance Date</u>
Select Project	1 June 1994
Complete design and specifications	1 September 1994
Commence construction	1 June 1995
Achieve full compliance	1 October 1995

In addition, the C&D Order prohibited any additional releases to the District's sewerage system by dischargers who did not discharge to the system prior to 24 January 1994, or any other increases in the volume of existing discharges.

6. Due to difficulties obtaining sufficient funding for the project, the Discharger was unable to comply with the compliance schedule in Finding No. 5 above. Funding was eventually secured from the County of Yolo in the form of Community Block Grants. Construction management of the project was provided by the County of Yolo Planning and Public Works Department.
7. Expansion of the wastewater facilities through the installation of two additional ponds, Nos. 6 and 7, has been completed in an area where a temporary stabilization pond was previously constructed (phase I). These additional ponds provide an auxiliary capacity of approximately 14.5 acre-feet. Based upon a 31 August 1998 water balance prepared by the Discharger's consultant, up to 141,579 gallons per day (gpd) of domestic waste from the community of Madison and the Center may be adequately managed by the seven stabilization ponds for treatment and disposal.
8. According to a water balance and reports submitted by the Discharger, with the addition of Pond Nos. 6 and 7, the wastewater treatment and disposal facility has adequate capacity for the existing system flows as well as development of approximately 13 of the 46 available parcels. This approximation assumes a density of 3.3 persons per single-family residential unit, with a per capita wastewater volume flow rate of 137 gallons per capita per day. Therefore, an equivalent dwelling unit (edu) would contribute approximately 452 gpd to the wastewater treatment and disposal facility. Flow calculations for connections other than single-family residential dwellings would need to be reviewed by the Board to determine the impact on available capacity.
9. Full build-out of the community would require further expansion of the wastewater treatment and disposal facilities to accommodate flows of up to 156,317 gpd. To this effect, the Discharger has proposed the addition of a 10-acre spreading area to provide the increased capacity necessary to treat and dispose of the anticipated wastewater flow from the remaining undeveloped parcels (phase II).
10. The Discharger's ponds are in Section 27, T10N, R1W, MDB&M, with surface

- water drainage to South Fork Willow Slough, as shown in Attachment A, which is attached hereto and part of the Order by reference.
11. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.
 12. The beneficial uses of South Fork Willow Slough are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; navigation; ground water recharge; fresh water replenishment; hydropower generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.
 13. Beneficial uses of ground water are municipal, industrial, and agricultural supply.
 14. The Basin Plan encourages recycling.
 15. Acting as the lead agency in accordance with the California Environmental Quality Act (CEQA), on 12 May 1995 the Yolo County Community Development Agency (County) issued a consideration and approval of a Negative Declaration, Finding of No Significant Impact (FONSI) for phases I and II of the Discharger's wastewater pond expansion project. On 18 July 1995, a Notice of Determination was signed and filed by the County. The Regional Board considered the FONSI and Notice of Determination and agrees with the County that no significant impacts to water quality will occur.
 16. This discharge is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq., (hereafter Title 27). The exemption, pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing waste discharge requirements, and
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to Title 22, CCR, Division 4.5, Chapter 11, as a hazardous waste.
 17. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
 18. The Board, in a public meeting, heard and considered all comments pertaining to

the discharge.

IT IS HEREBY ORDERED that Order No. 89-235 is rescinded and Madison Community Services District, it's agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. The discharge of waste classified as 'hazardous', as defined in Sections 2521(a) of Title 23, CCR, Section 2510, et seq. (hereafter Chapter 15), or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.
4. The discharge of wastewater or recycled water to the proposed phase II land application area (Designated Disposal Area) is prohibited until approved by the Executive Officer.

B. Discharge Specifications:

1. The monthly average dry weather discharge flow shall not exceed 141,579 gpd until the phase II designated land application area has been approved and implemented. Upon completion and approval of phase II, the average dry weather discharge shall not exceed 156,317 gpd.
2. New residential service connections shall be based upon a design flow of 452 gpd/edu. Other types of service connection design flows shall be reviewed by the Board to determine impacts on available capacity.
3. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater collection, treatment and disposal area, or property owned by the Discharger.
4. As means of discerning compliance with Discharge Specification No. B.3, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
5. The ponds shall not have a pH of less than 6.5 or greater than 8.5.

6. A 2.0-foot freeboard shall be maintained in the ponds at all times.
7. As a means of discerning compliance with Discharge Specification No. B.6, the Discharger shall provide a staff gauge and survey mechanism for each pond.
8. On or about **15 October of each year**, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification No. B.6.
9. The ponds and Designated Disposal Area shall be managed to prevent breeding of mosquitoes. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
 - d. There shall be no standing water in the Designated Disposal Area 48 hours after wastewater or recycled water is applied.
10. Public contact with wastewater and recycled water shall be precluded through such means as fences, signs, and other acceptable alternatives.
11. The ponds and Designated Disposal Area shall have sufficient capacity to accommodate allowable wastewater flow, design seasonal precipitation and ancillary inflow and infiltration during the non-irrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed in accordance with historical rainfall patterns.
12. The discharge shall remain within designated treatment and disposal areas at all times.
13. Recycled wastewater shall meet the criteria contained in Title 22, Division 4, California Administrative Code.
14. The effluent from the treatment ponds to the Designated Disposal Area shall not exceed the following limits:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Monthly Median</u>	<u>Daily Maximum</u>
BOD ₅ ¹	mg/l	40		80
Settleable Solids	ml/l/hr	0.2		0.5
Suspended Solids	mg/l	30		60
Coliform	MPN/100ml		23	240

¹Five-day, 20°C biochemical oxygen demand (BOD).

15. The discharge of recycled water shall not occur offsite of the Dischargers treatment and disposal facility.
16. The discharge shall not cause degradation of any water supply.

C. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste, as set forth in CCR, Title 27, Division 2, Subdivision 1, Section 20005, et seq. and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice from a previously approved practice shall be reported to the Executive Officer and U.S. Environmental Protection Agency (EPA) Regional Administrator at least 90 days in advance of the change.
3. Use and disposal of sewage shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

If the State Water Resources Control Board and the Regional Water Quality Control Boards are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

D. Groundwater Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except for coliform. For coliform, increases shall not cause the most probable number of total coliform organisms to exceed 2.2/100 ml over any 7-day period.

E. Provisions:

1. The Discharger shall comply with the attached Monitoring and Reporting Program No. 5-00-039, which is part of this order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. In the event of any change in control of ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
4. At least **90 days** prior to termination or expiration of any lease, contract, or agreement involving disposal or recycling areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
5. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Board or court orders requiring corrective action or imposing civil liability, or in revision or rescission of this order.
6. A copy of this order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
7. The Board will review this Order periodically and may revise requirements

when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 17 March 2000.

GARY M. CARLTON, Executive Officer

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17 March 2000
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